Xác định GPIO: dùng chân GPIO 60 (chân P9_12 trên BBB).

Kích hoạt GPIO

```
Welcome to Buildroot
buildroot login: root
# echo 60 > /sys/class/gpio/export
#
```

Cấu hình chiều (Direction)

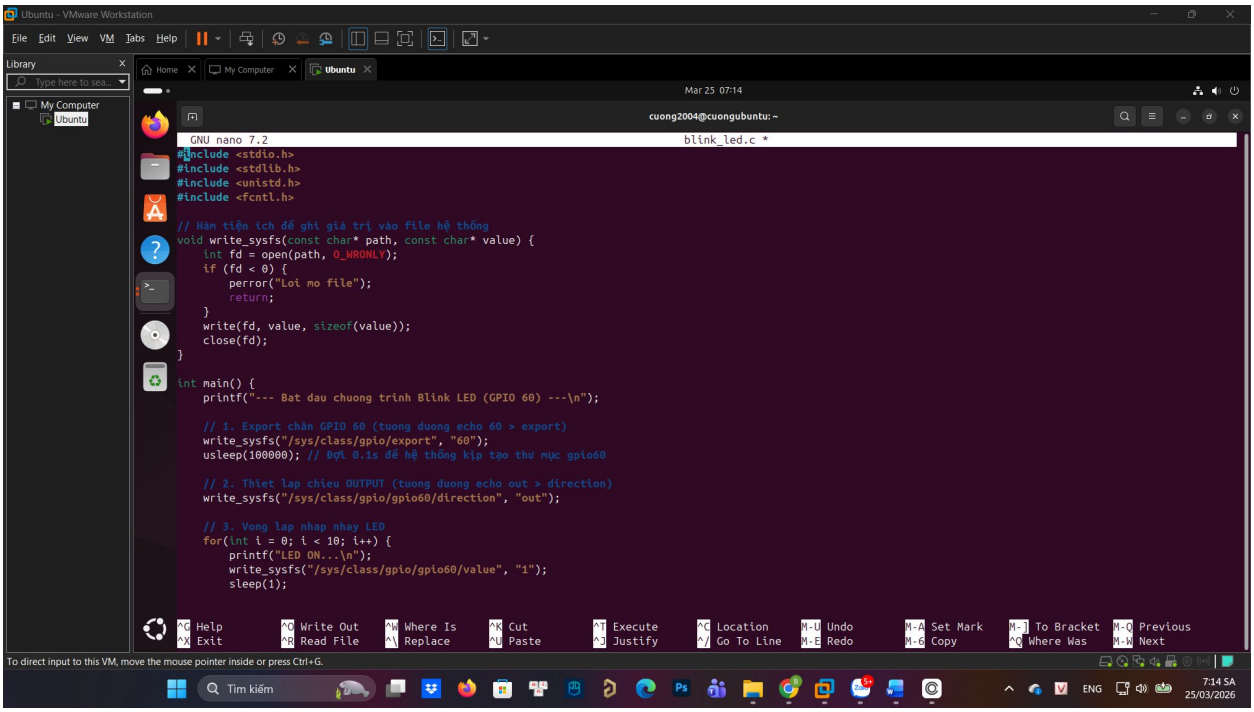
```
# echo out > /sys/class/gpio/gpio60/direction
#
```

Điều khiển led:

Bật :
```
# echo 1 > /sys/class/gpio/gpio60/value
```

Tắt :
```
# echo 0 > /sys/class/gpio/gpio60/value
```

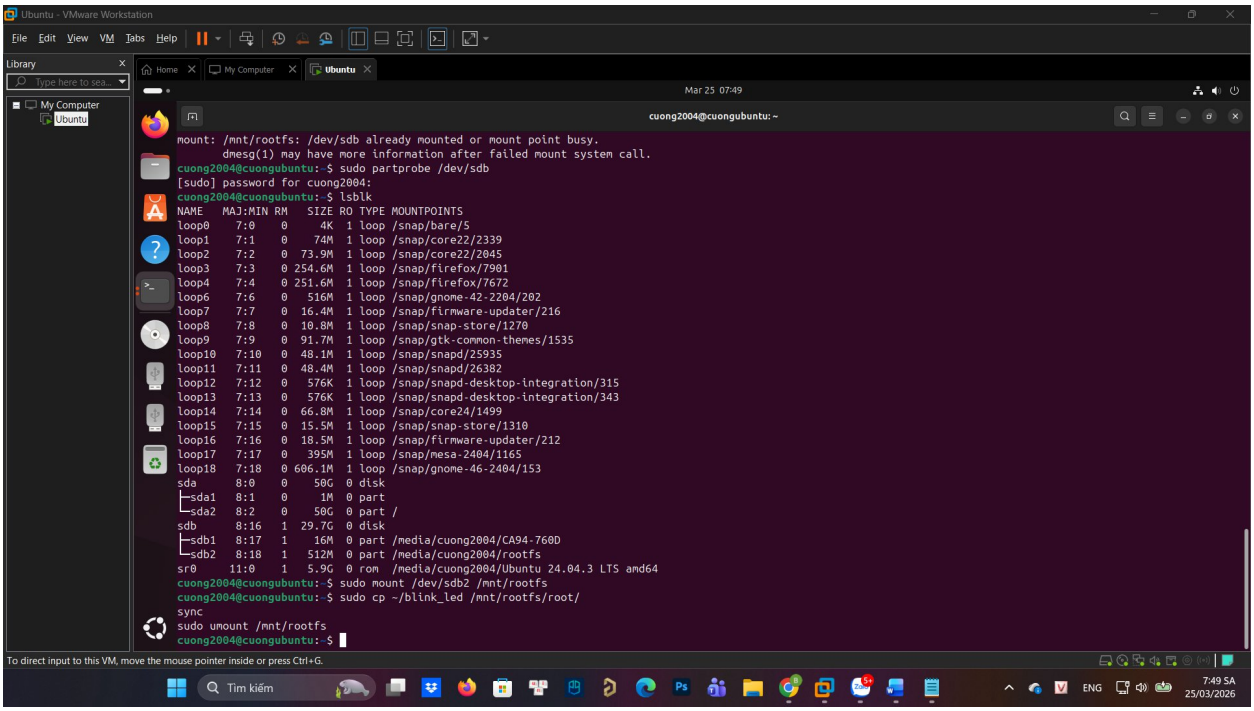
Tạo file blink_led.c



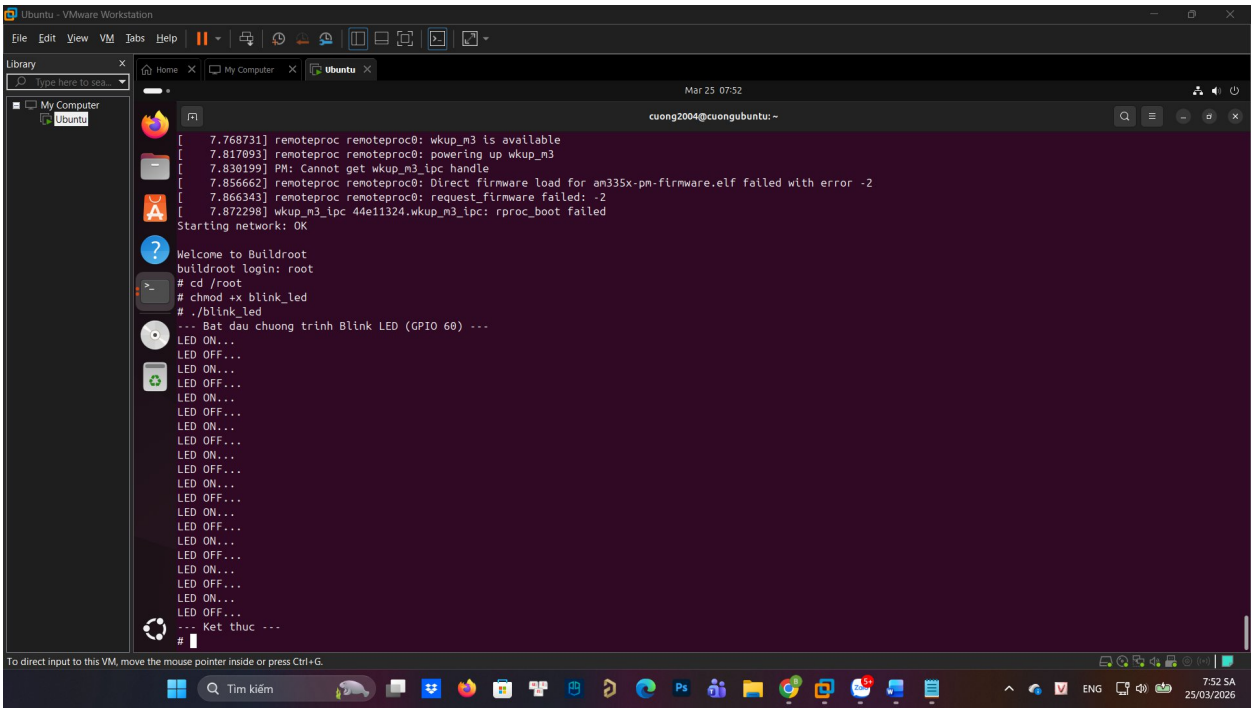
Biên dịch chéo:

```
cuong2004@cuongubuntu:~$ /home/cuong2004/buildroot/output/host/bin/arm-linux-gcc blink_led.c -o blink_led
```

Đưa file blink_led vào mạch qua thẻ nhớ:



Chạy Chtrinh:

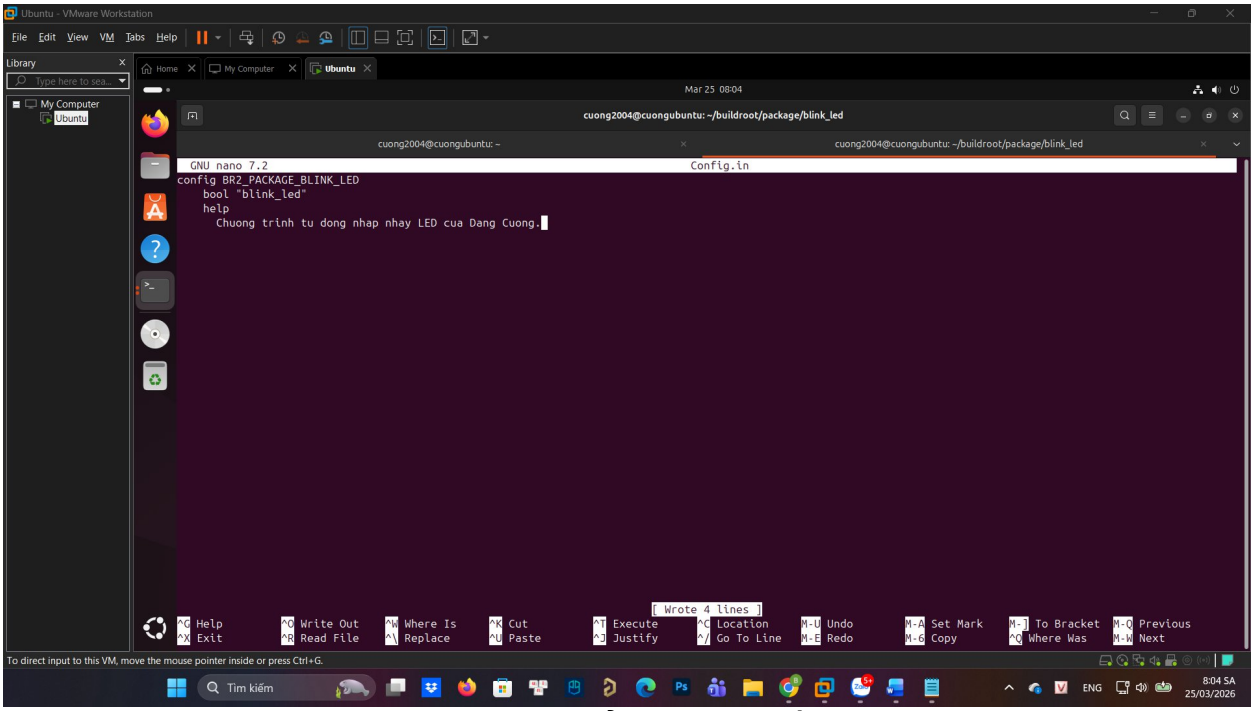


Đóng gói vào Buildroot:

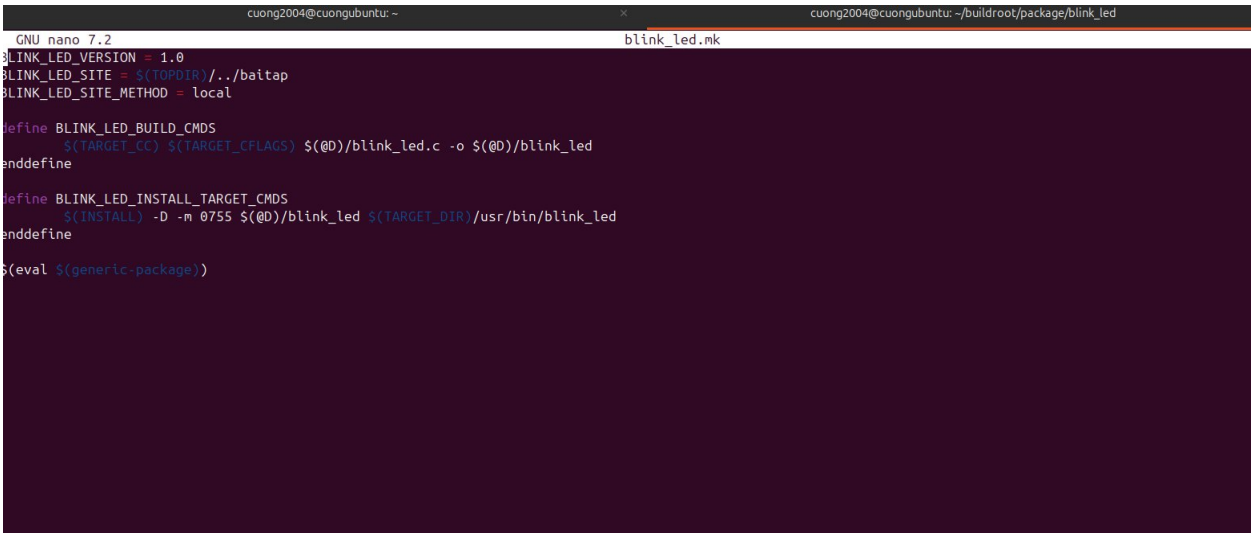
Tạo cấu trúc thư mục

```
cuong2004@cuongubuntu:~$ mkdir -p ~/buildroot/package/blink_led
```

Tạo file package/blink_led/Config.in (để nó hiện trong make menuconfig)



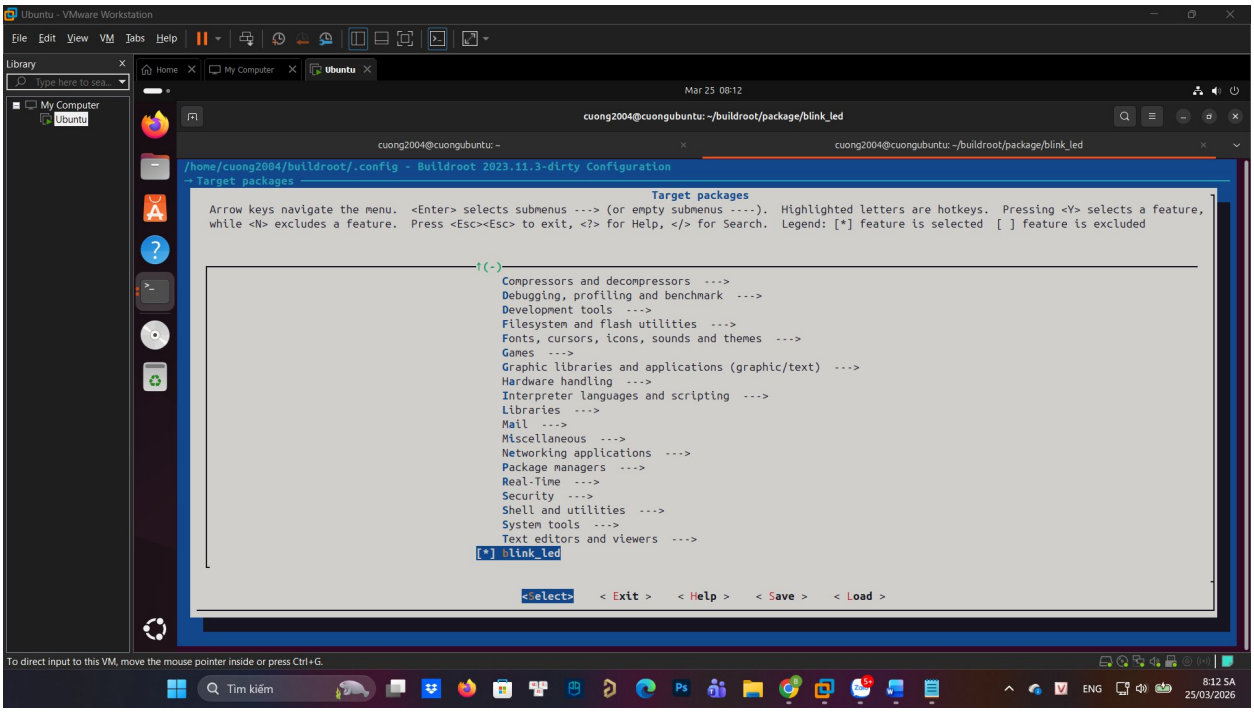
Tạo file package/blink_led/blink_led.mk (Để Buildroot biết cách build)



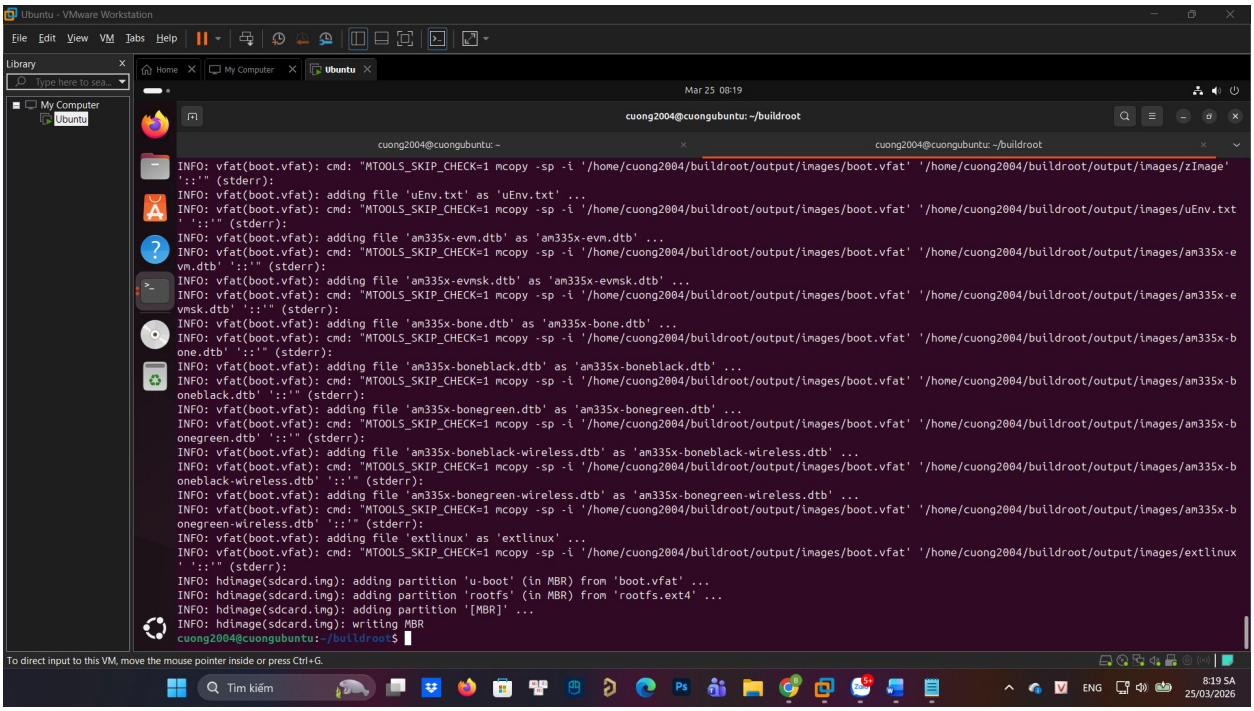
Kết nối với buildroot tổng

```
ink_led$ nano ~/buildroot/package/Config.in
ink_led$
        source "package/most/Config.in"
        source "package/nano/Config.in"
        source "package/uemacs/Config.in"
        source "package/vim/Config.in"
endmenu
source "package/blink_led/Config.in"
endmenu
```

Check trong menuconfig thì hiện file blink_led



Biên dịch toàn bộ Image



Check xem file đã nằm đúng chỗ chưa

```
cuong2004@cuongubuntu:~/buildroot$ ls -lh ~/buildroot/output/images/sdcard.img
-rw-r--r-- 1 cuong2004 cuong2004 77M Mar 25 08:19 /home/cuong2004/buildroot/output/images/sdcard.img
cuong2004@cuongubuntu:~/buildroot$
```

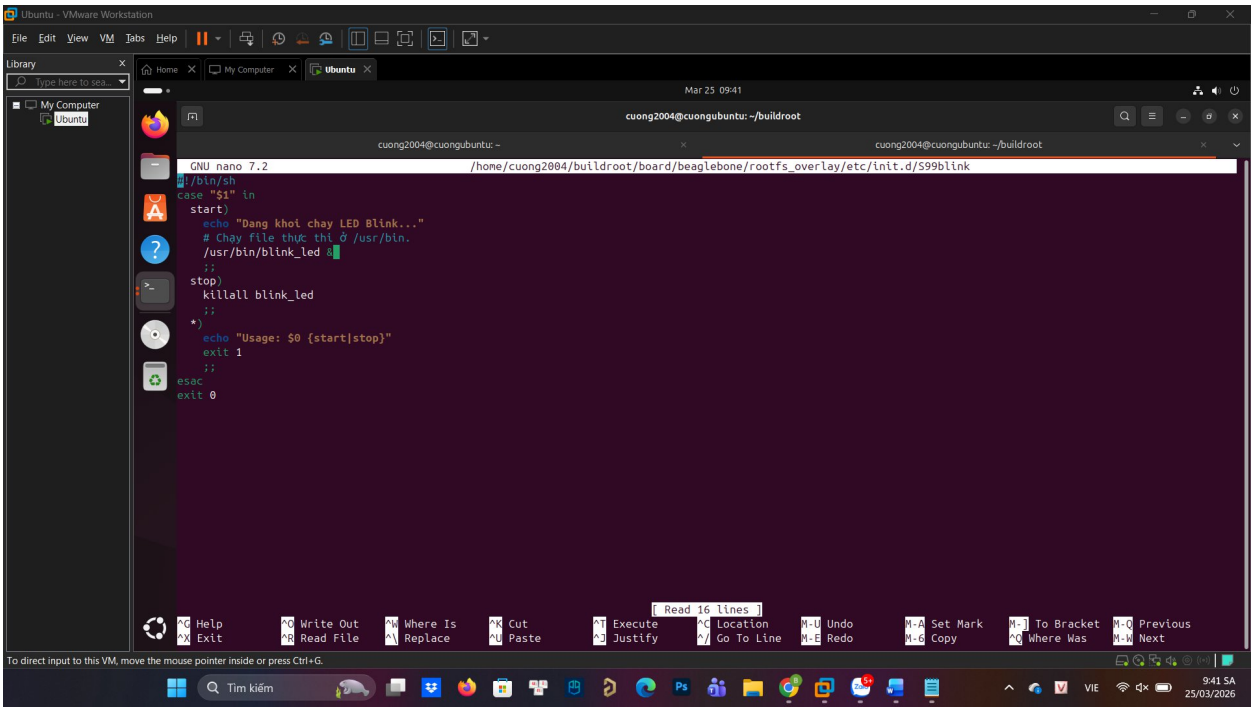
Tự động hóa

Tạo thư mục chứa overlay

```
cuong2004@cuongubuntu:~/buildroot$ mkdir -p ~/buildroot/board/beaglebone/rootfs_overlay/etc/init.d/
cuong2004@cuongubuntu:~/buildroot$
```

Tạo Script khởi động

```
cuong2004@cuongubuntu:~/buildroot$ nano ~/buildroot/board/beaglebone/rootfs_overlay/etc/init.d/S99blink
```



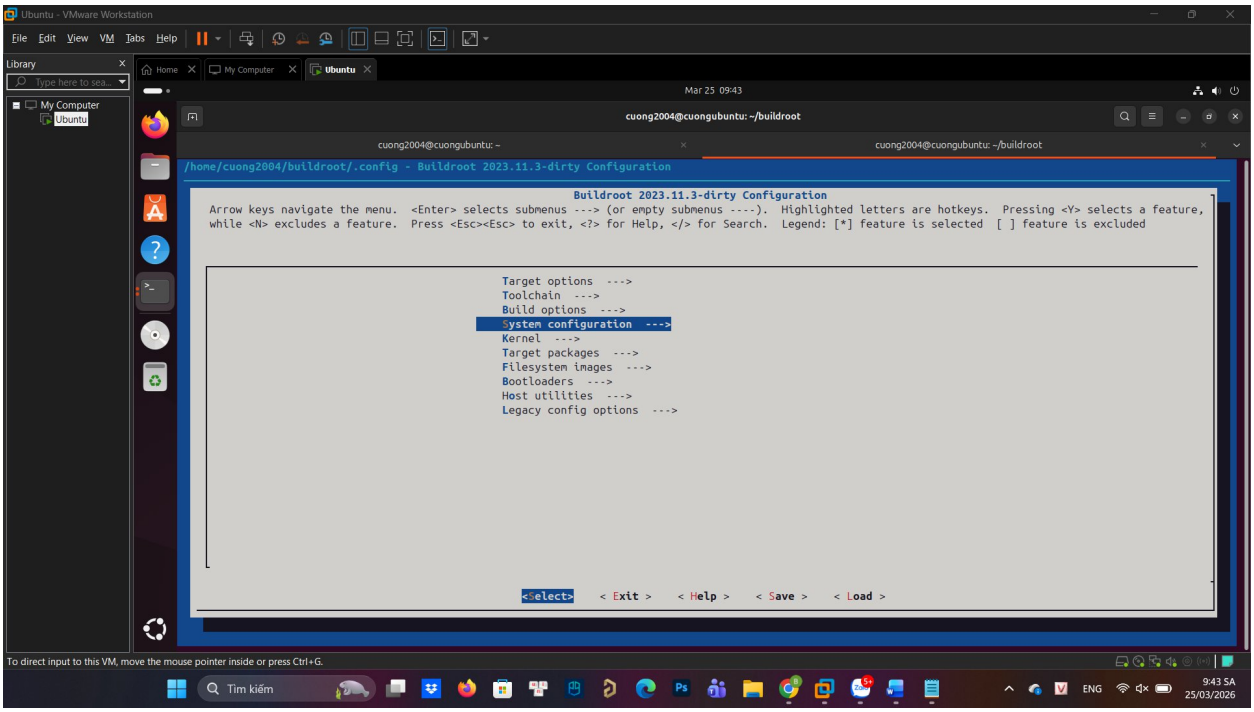
Cấp quyền thi hành (execute)

```
cuong2004@cuongubuntu:~/buildroot$ chmod +x ~/buildroot/board/beaglebone/rootfs_overlay/etc/init.d/S99blink
```

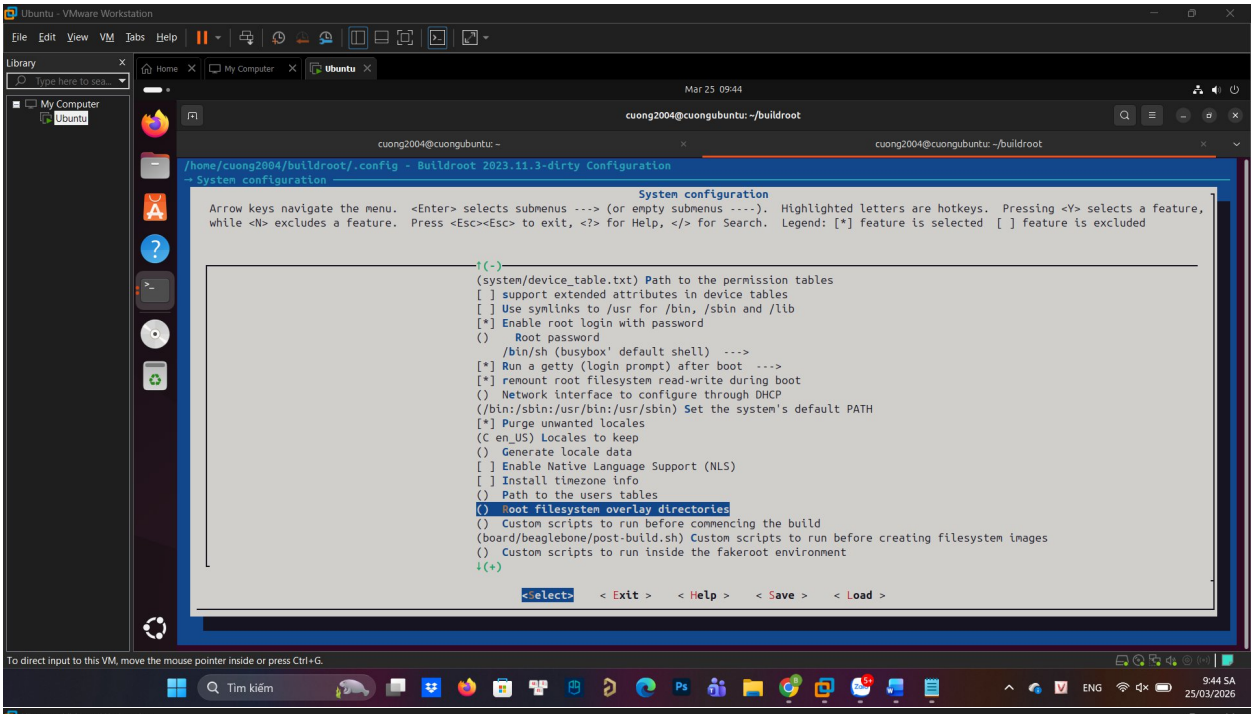
khai báo buildroot

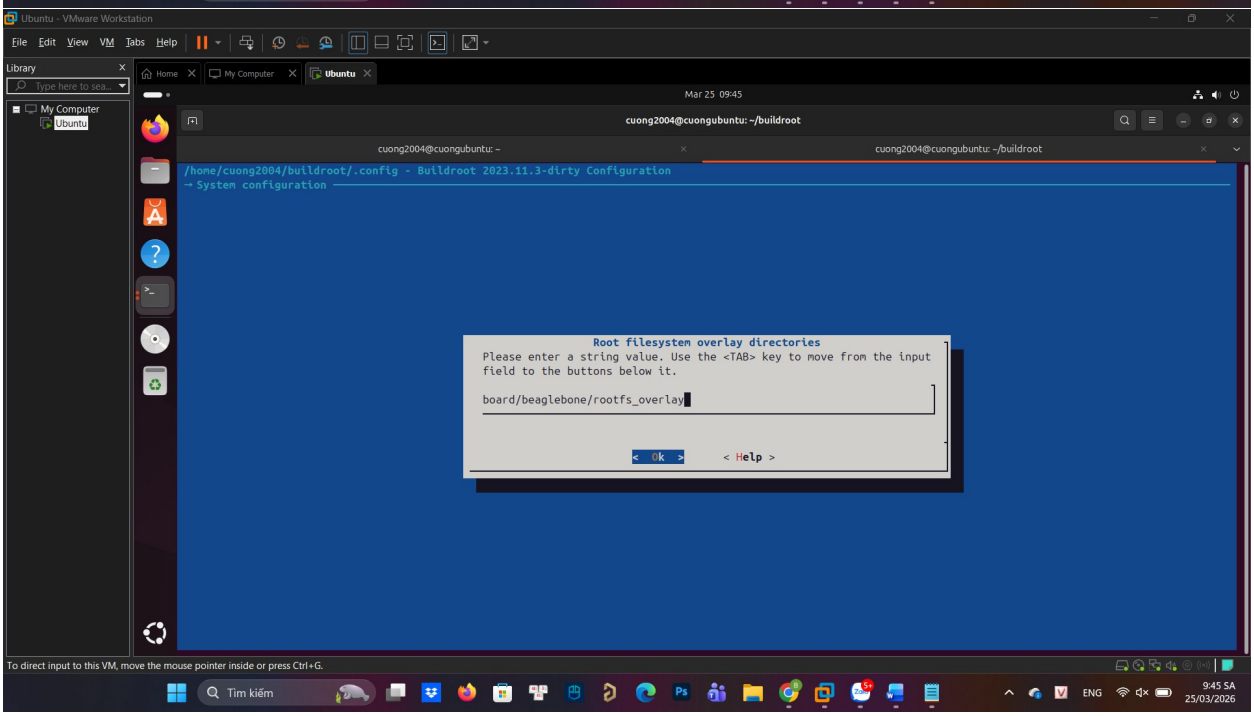
Vào menuconfig

Đến system configuration

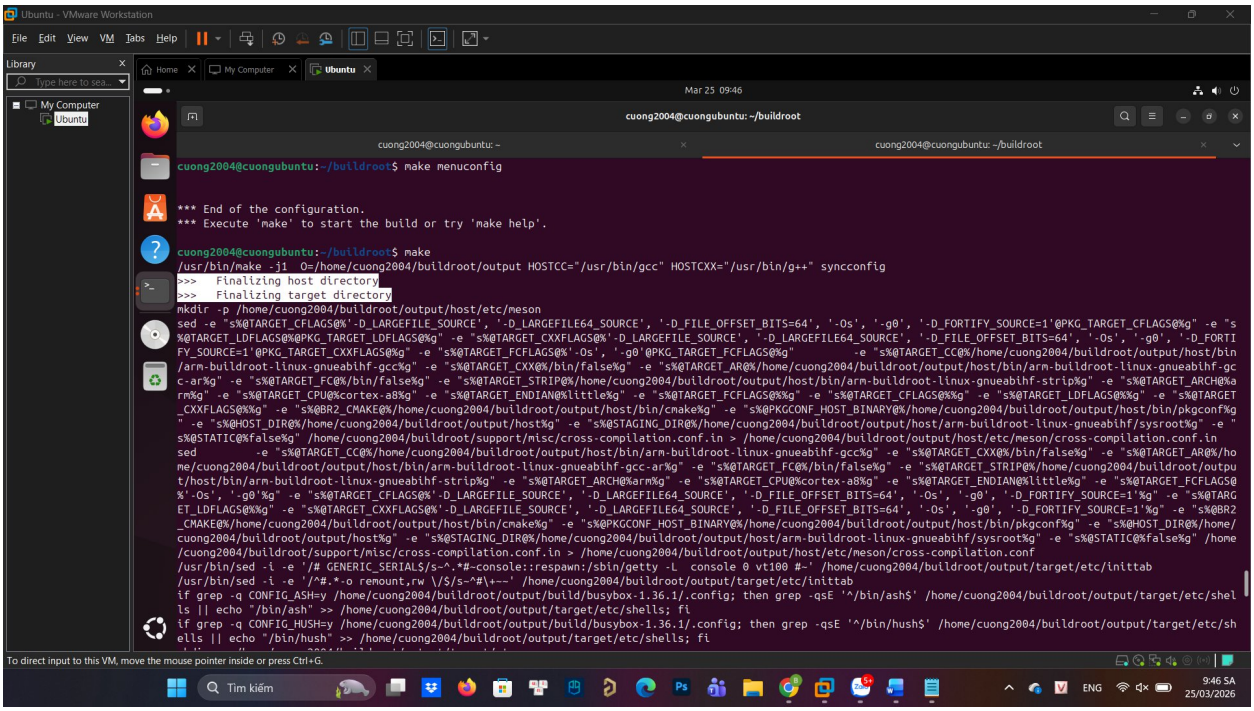


Vào root filesystem overlay directories



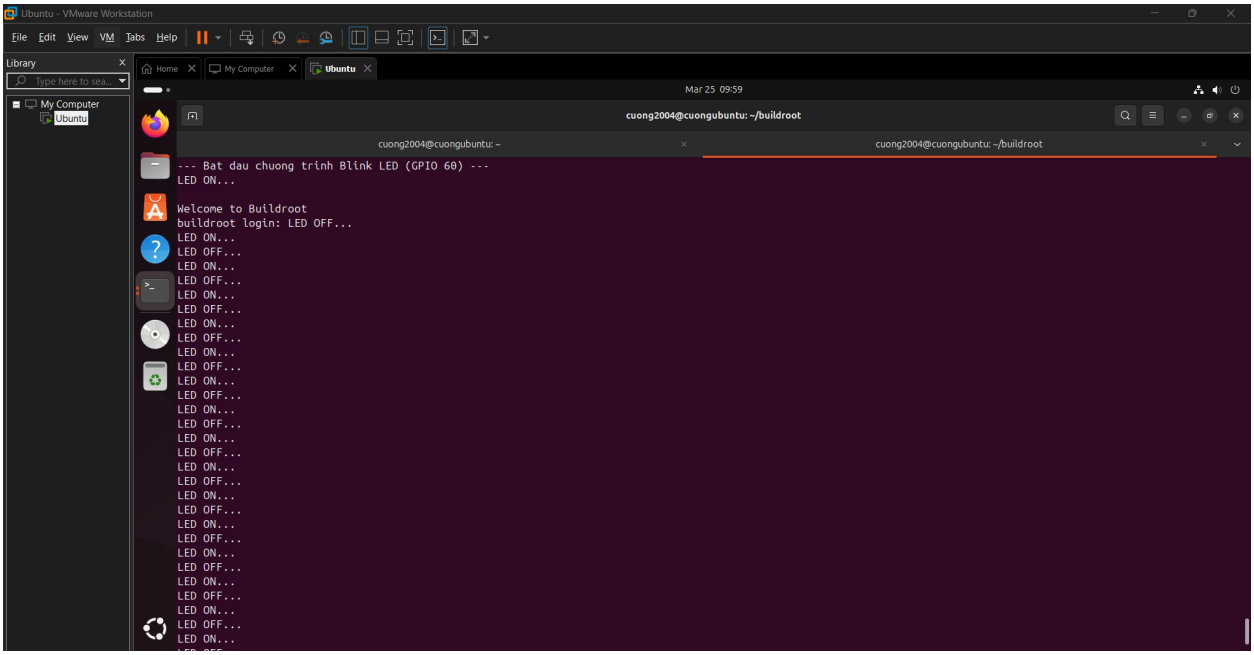


Biên dịch lại



Flash thẻ nhớ

```
cuong2004@cuongubuntu:~/buildroot$ sudo dd if=~/buildroot/output/images/sdcard.img of=/dev/sdb bs=1M status=progress
sync
[sudo] password for cuong2004:
76+1 records in
76+1 records out
79692288 bytes (80 MB, 76 MiB) copied, 0.118964 s, 670 MB/s
```

lệnh dd để ghi trực tiếp file ảnh hệ thống (Image) xuống thẻ đảm bảo cấu trúc phân vùng Boot và RootFS được giữ nguyên

cắm thẻ nhớ vào BBB